

MASTER APP BUILD SPECIFICATION

Product Name

TranquilityOS

Tagline

The AI-Powered Operating System for Service Businesses

Product Vision

Build a modern business-management web application that allows service businesses to manage their leads, customers, quotations, invoices, jobs, appointments, payments, employees, documents, and customer communication from one platform.

The application should also include an AI assistant that can generate quotations, summarize customer activity, draft messages, create follow-ups, answer questions about the business, and help users manage daily operations.

The first version should target:

- Aluminum window and door companies
- Electricians
- Plumbers
- General contractors
- HVAC companies
- Roofers
- Painters
- Landscapers
- Cleaning companies
- Appliance repair companies
- Other home-service businesses

Do not build every planned feature at once. Build the application in clearly separated phases, beginning with the MVP.

1. PRODUCT OBJECTIVES

The application must help service businesses:

1. Capture and manage leads.

2. Store customer information.
 3. Create professional quotations.
 4. Convert quotations into jobs.
 5. Schedule estimates and installations.
 6. Create and send invoices.
 7. Record payments and outstanding balances.
 8. Assign jobs to employees.
 9. Store photos, documents, measurements, and notes.
 10. Communicate with customers.
 11. Automate follow-ups and reminders.
 12. Use AI to complete routine business tasks.
 13. Track revenue, jobs, quotations, invoices, and performance.
 14. Provide customers with an online portal.
 15. Operate from desktop, tablet, and mobile devices.
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2. USER TYPES

The system should support the following user roles.

Platform Owner

The owner of TranquilityOS.

Permissions:

- View all registered businesses
- Manage subscription plans
- Suspend or activate accounts
- View platform revenue
- View AI usage
- Manage application-wide settings
- Manage support requests
- View system health
- Manage feature access
- Manage industries and templates

Business Owner

The owner of a company using TranquilityOS.

Permissions:

- Full access to the business account
- Add employees

- Manage roles and permissions
- View financial reports
- Create quotations and invoices
- Manage customers and jobs
- Manage settings
- Configure AI
- Configure automations
- Manage subscription and billing

Manager

Permissions:

- Manage leads
- Manage customers
- Manage quotations
- Manage jobs
- Assign employees
- Manage the calendar
- View selected reports
- Communicate with customers

Managers should not automatically have access to subscription billing or highly sensitive company settings.

Sales Representative

Permissions:

- Add and manage leads
- Add customers
- Create quotations
- Schedule estimates
- Add notes
- Follow up prospects
- View assigned leads and customers

Technician or Field Worker

Permissions:

- View assigned jobs
- View customer address and contact details
- Update job status
- Upload job photos
- Add job notes
- Complete checklists
- Record materials used

- Capture customer signatures
- Mark jobs complete

Accountant

Permissions:

- View quotations
- View invoices
- Record payments
- Track expenses
- View financial reports
- Export financial data

Customer

Customers use the customer portal.

Permissions:

- View and approve quotations
- View scheduled appointments
- Track job progress
- Pay invoices
- Download receipts
- Upload documents or photos
- Message the business
- Submit warranty requests
- Leave a review

3. CORE MVP MODULES

Build the following modules first.

Module 1: Authentication and Onboarding

Features:

- User registration
- Login
- Logout
- Password reset
- Email verification
- Secure sessions

- Business account creation
- Business profile setup
- Industry selection
- Currency selection
- Time-zone selection
- Tax settings
- Logo upload
- Company contact details
- Default quotation terms
- Default invoice terms
- Team-member invitations
- Role selection

Business profile fields:

- Business name
- Business logo
- Business email
- Telephone number
- WhatsApp number
- Website
- Street address
- City
- Parish, state, or region
- Country
- Currency
- Time zone
- Tax number
- Registration number
- Default deposit percentage
- Default payment terms
- Default quotation validity period

The application must support multiple businesses. Every business must only see its own data.

Module 2: Main Dashboard

The dashboard should display:

- Revenue this month
- Payments received
- Outstanding invoice balance
- Number of active jobs
- Jobs scheduled today
- New leads
- Quotations awaiting approval

- Approved quotations
- Overdue invoices
- Recent customer activity
- Upcoming appointments
- Employee activity
- Sales pipeline
- Monthly revenue chart
- Quotation conversion rate
- AI recommendations

Dashboard actions:

- Add lead
- Add customer
- Create quotation
- Create invoice
- Create job
- Schedule appointment
- Record payment
- Ask AI assistant

The dashboard must be responsive and easy to use on mobile devices.

Module 3: Lead Management

Lead fields:

- Lead name
- Company name
- Telephone number
- WhatsApp number
- Email
- Address
- Service requested
- Lead source
- Estimated value
- Priority
- Assigned employee
- Status
- Notes
- Next follow-up date
- Date created
- Last contact date

Lead statuses:

- New

- Contacted
- Qualified
- Estimate scheduled
- Quotation sent
- Negotiation
- Won
- Lost

Lead sources:

- Website
- Facebook
- Instagram
- WhatsApp
- Telephone
- Referral
- Walk-in
- Google
- Advertising
- Other

Lead features:

- Search
- Filter
- Sort
- Assign
- Add notes
- Add tasks
- Schedule follow-up
- Convert lead to customer
- Convert lead to quotation
- View activity timeline
- View communication history
- Bulk actions
- Import leads from CSV
- Export leads to CSV

Module 4: Customer Relationship Management

Customer fields:

- Full name
- Business name
- Telephone number
- WhatsApp number

- Email
- Billing address
- Service address
- GPS location
- Customer type
- Assigned representative
- Tax number
- Notes
- Tags
- Date created

Customer profile sections:

- Overview
- Contact information
- Leads
- Quotations
- Jobs
- Appointments
- Invoices
- Payments
- Documents
- Photos
- Messages
- Notes
- Activity timeline
- Warranty information

Customer types:

- Residential
- Commercial
- Government
- Church
- School
- Contractor
- Property manager
- Other

Features:

- Add customer
- Edit customer
- Archive customer
- Search customers
- Filter customers
- Add tags
- Upload documents

- Upload photos
 - Add internal notes
 - View outstanding balance
 - View total customer value
 - Send email
 - Send WhatsApp message through an integration
 - Create quotation
 - Create invoice
 - Create job
 - Schedule appointment
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Module 5: Products and Services

Create a reusable product and service catalogue.

Fields:

- Name
- Category
- Description
- Unit
- Cost price
- Selling price
- Taxable status
- Active or inactive
- Internal code
- Image
- Default labor cost
- Default material cost

Example units:

- Each
- Square foot
- Linear foot
- Hour
- Day
- Job
- Set
- Length
- Meter
- Bundle

Example categories:

- Windows

- Doors
- Electrical installation
- Plumbing
- Labor
- Materials
- Consultation
- Delivery
- Installation
- Repair
- Maintenance

Users must be able to add custom products and services.

Module 6: Quotation and Estimate Builder

Users must be able to create professional quotations.

Quotation fields:

- Quotation number
- Customer
- Customer address
- Service location
- Date created
- Expiration date
- Sales representative
- Line items
- Quantity
- Unit
- Unit price
- Discount
- Tax
- Deposit
- Subtotal
- Total
- Notes
- Terms and conditions
- Internal notes
- Status

Quotation statuses:

- Draft
- Sent
- Viewed
- Approved

- Rejected
- Expired
- Converted
- Cancelled

Quotation features:

- Create manually
- Create from product catalogue
- Duplicate quotation
- Add custom line item
- Add measurements
- Add materials
- Add labor
- Add photos
- Add customer signature
- Add company logo
- Calculate totals automatically
- Apply percentage or fixed discount
- Apply tax
- Apply deposit percentage
- Add payment schedule
- Preview quotation
- Generate PDF
- Email quotation
- Share quotation link
- Send through WhatsApp
- Customer online approval
- Customer rejection with reason
- Convert approved quotation to job
- Convert approved quotation to invoice
- Record quotation activity

Quotation numbering example:

TOS-Q-000001

Each business should be able to configure its document prefix.

Module 7: AI Quotation Generator

The user should be able to enter a written or spoken description such as:

“Create a quotation for Mr. Brown for three white sliding windows. Two windows are 48 inches by 48 inches and one is 72 inches by 48 inches. Charge \$35,000 for each smaller window and \$42,000 for the larger window. Require a 70 percent deposit.”

The AI should:

1. Identify the customer.
2. Identify products or services.
3. Extract measurements.
4. Extract quantities.
5. Extract prices.
6. Extract colors or specifications.
7. Apply deposit requirements.
8. Create a draft quotation.
9. Ask the user to review the quotation before it is sent.
10. Never send or approve financial documents without user confirmation.

AI-generated quotations must be saved as drafts.

Module 8: Job Management

Jobs can be created manually or from approved quotations.

Job fields:

- Job number
- Customer
- Service address
- Linked quotation
- Linked invoice
- Job title
- Description
- Assigned team
- Start date
- Due date
- Priority
- Job status
- Estimated cost
- Estimated revenue
- Actual cost
- Job notes
- Customer instructions
- Internal instructions
- Materials required
- Photos
- Documents
- Checklist
- Customer signature
- Completion date

Job statuses:

- New
- Awaiting deposit
- Scheduled
- In progress
- On hold
- Awaiting materials
- Awaiting customer
- Completed
- Cancelled
- Warranty service

Job features:

- Assign one or more employees
- Add tasks
- Add checklists
- Upload before-and-after photos
- Upload documents
- Record measurements
- Record materials used
- Record job expenses
- Add progress updates
- Capture customer signature
- Mark job completed
- Generate completion report
- Notify customer of status changes
- View job profitability
- View job activity timeline

Module 9: Calendar and Scheduling

Calendar views:

- Day
- Week
- Month
- Employee
- Job
- Appointment

Appointment types:

- Consultation
- Site visit

- Measurement
- Estimate
- Installation
- Repair
- Maintenance
- Follow-up
- Meeting
- Other

Appointment fields:

- Title
- Customer
- Job
- Address
- Start date and time
- End date and time
- Assigned employee
- Notes
- Reminder
- Status

Appointment statuses:

- Scheduled
- Confirmed
- In progress
- Completed
- Cancelled
- Rescheduled
- No-show

Features:

- Drag-and-drop scheduling
 - Detect employee scheduling conflicts
 - Send customer reminders
 - Send employee reminders
 - Add appointment from customer profile
 - Add appointment from lead
 - Add appointment from job
 - Sync with an external calendar in a later phase
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Module 10: Invoice Management

Invoice fields:

- Invoice number
- Customer
- Linked quotation
- Linked job
- Invoice date
- Due date
- Line items
- Quantity
- Unit price
- Discount
- Tax
- Deposit paid
- Subtotal
- Total
- Amount paid
- Balance
- Notes
- Payment instructions
- Status

Invoice statuses:

- Draft
- Sent
- Viewed
- Partially paid
- Paid
- Overdue
- Cancelled
- Refunded

Invoice features:

- Create from quotation
- Create from job
- Create manually
- Duplicate invoice
- Add custom line items
- Add discounts
- Add tax
- Record deposits
- Record partial payments
- Apply late fee
- Generate PDF

- Email invoice
- Send invoice link
- Share through WhatsApp
- Download invoice
- Generate receipt
- Send payment reminder
- Show payment history

Invoice numbering example:

TOS-I-000001

Module 11: Payments and Receipts

Payment fields:

- Customer
- Invoice
- Amount
- Currency
- Payment date
- Payment method
- Transaction reference
- Notes
- Recorded by
- Receipt number

Payment methods:

- Cash
- Bank transfer
- Credit card
- Debit card
- PayPal
- Stripe
- Mobile money
- Cheque
- Other

Features:

- Record payment
- Record partial payment
- Apply payment to invoice
- Generate receipt
- Email receipt

- Download receipt PDF
- Refund payment
- View payment history
- Update invoice balance automatically

Online payment integration should be added after the basic manual payment system is stable.

Module 12: Employee and Team Management

Employee fields:

- Full name
- Email
- Telephone
- Role
- Job title
- Profile photo
- Status
- Hire date
- Hourly rate
- Permissions
- Assigned jobs
- Emergency contact
- Notes

Features:

- Invite employee
- Deactivate employee
- Assign roles
- Configure permissions
- Assign jobs
- View employee calendar
- View employee activity
- Record time entries
- Upload employee documents
- View completed jobs

Do not expose sensitive financial information to employees unless their role permits it.

Module 13: Tasks and Checklists

Task fields:

- Title

- Description
- Assigned user
- Customer
- Lead
- Job
- Due date
- Priority
- Status
- Reminder
- Notes

Task statuses:

- Not started
- In progress
- Completed
- Cancelled

Features:

- Create reusable checklist templates
- Attach checklist to a job
- Require specific checklist items before job completion
- Add photos to checklist items
- Assign checklist items to employees

Module 14: Documents and Media

Support uploads for:

- Images
- PDF files
- Contracts
- Drawings
- Measurements
- Receipts
- Warranty documents
- Identification documents
- Job completion reports

Features:

- Upload
- Preview
- Download
- Rename

- Categorize
- Delete
- Search
- Attach to customer
- Attach to lead
- Attach to quotation
- Attach to invoice
- Attach to job
- Record uploader and upload date

Files must be stored securely. Businesses must not be able to access another business's files.

Module 15: Communication Center

Store customer communication in one timeline.

Communication types:

- Email
- SMS
- WhatsApp
- Telephone call
- Internal note
- Website chat
- AI conversation

Features:

- Draft email
- Send email
- Save email template
- Draft WhatsApp message
- Log telephone call
- Add call outcome
- Schedule follow-up
- View communication history
- Attach communication to lead, customer, quotation, invoice, or job

Initial MVP can support email and manually logged calls. WhatsApp and SMS integrations can be added afterward.

Module 16: AI Business Assistant

Create a chat interface inside the application.

Example commands:

- "Show me all overdue invoices."
- "Which quotations are waiting for approval?"
- "Create a follow-up message for Mr. Brown."
- "Summarize this customer."
- "How much revenue did we make this month?"
- "Show jobs scheduled for tomorrow."
- "Create a quotation for two sliding windows."
- "Write a payment reminder."
- "Which leads have not been contacted?"
- "What tasks are overdue?"

AI assistant rules:

1. The AI must only access data belonging to the current business.
2. The AI must respect user permissions.
3. The AI may prepare drafts.
4. The AI must not send messages, delete records, issue refunds, approve quotations, or perform financial actions without user confirmation.
5. All AI actions must be logged.
6. The user should be able to review AI-generated content.
7. The AI should explain when it does not have enough information.
8. The AI must not invent customer, pricing, payment, or job information.

Module 17: Reports and Analytics

MVP reports:

- Revenue by month
- Payments received
- Outstanding invoices
- Overdue invoices
- Quotation value
- Quotation approval rate
- Sales by employee
- Leads by source
- Lead conversion rate
- Jobs by status
- Jobs completed
- Revenue by service
- Top customers
- Customer balances

Features:

- Date filtering
- Employee filtering
- Customer filtering
- Export to CSV
- Export to PDF
- Dashboard charts
- Printable reports

Module 18: Customer Portal

Customers should receive a secure link or login.

Customer portal pages:

- Dashboard
- Quotations
- Invoices
- Payments
- Jobs
- Appointments
- Documents
- Messages
- Warranty requests
- Profile

Customer actions:

- View quotation
 - Approve quotation
 - Reject quotation
 - Add comment
 - Download PDF
 - View invoice
 - Pay invoice in a later phase
 - Download receipt
 - View job progress
 - Upload photos
 - Send message
 - Request service
 - Submit warranty claim
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4. NAVIGATION STRUCTURE

Use a left sidebar on desktop and a mobile menu on smaller screens.

Main navigation:

1. Dashboard
2. Leads
3. Customers
4. Quotations
5. Jobs
6. Calendar
7. Invoices
8. Payments
9. Products and Services
10. Team
11. Tasks
12. Reports
13. AI Assistant
14. Documents
15. Settings

Platform administration should use a separate admin interface.

5. REQUIRED WORKFLOWS

Lead-to-Customer Workflow

New lead

- Contact lead
- Qualify lead
- Schedule estimate
- Create quotation
- Customer approves
- Convert to customer
- Create job
- Create invoice
- Record payment
- Complete job
- Request review

Quotation Workflow

Draft

- Review
- Send
- Customer views
- Customer approves or rejects
- Convert approved quotation into job and invoice

Job Workflow

New

- Awaiting deposit
- Scheduled
- In progress
- Completed
- Customer signature
- Final invoice
- Payment
- Review request

Invoice Workflow

Draft

- Send
- Viewed
- Partial payment or full payment
- Paid

When due date passes and a balance remains:

Invoice becomes overdue

- Reminder is generated
- Late fee may be applied based on business settings

Customer Follow-Up Workflow

Quotation sent

- No reply after configured period
 - Generate follow-up message
 - Notify assigned sales representative
 - Record follow-up activity
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6. AUTOMATION ENGINE

Build a basic rule-based automation system after the core MVP is working.

Automation structure:

Trigger → Condition → Action

Example triggers:

- Lead created
- Quotation sent
- Quotation approved
- Quotation expired
- Invoice sent
- Invoice overdue
- Payment received
- Appointment scheduled
- Job completed
- Customer inactive
- Task overdue

Example actions:

- Send email
- Create task
- Notify employee
- Update status
- Generate follow-up message
- Schedule reminder
- Request review
- Create invoice
- Create job

Examples:

- When a quotation is approved, create a job.
- When an invoice is overdue by three days, create a reminder task.
- When a job is completed, send a review request.
- When a lead has not been contacted in one day, notify the assigned employee.
- When a deposit is recorded, update the job to "Scheduled."

The first automation version should provide prebuilt templates rather than a complex visual builder.

7. DATABASE STRUCTURE

Create a relational database with the following primary tables:

- users
- businesses
- business_users
- roles
- permissions
- subscriptions
- leads
- lead_activities
- customers
- customer_addresses
- customer_notes
- products
- product_categories
- quotations
- quotation_items
- quotation_activities
- jobs
- job_assignments
- job_tasks
- job_checklists
- job_checklist_items
- job_materials
- job_expenses
- appointments
- invoices
- invoice_items
- payments
- receipts
- employees
- employee_time_entries
- tasks
- files
- messages
- message_templates
- notifications
- automation_rules
- automation_logs
- ai_conversations
- ai_messages
- ai_actions
- audit_logs
- business_settings

Every business-owned table must include a business identifier.

Use database constraints, validation, and secure authorization to prevent cross-business access.

8. MULTI-TENANT REQUIREMENTS

The application is a multi-tenant SaaS platform.

Requirements:

- Each business has a separate workspace.
- Users may belong to one or more businesses.
- A user must select the active business.
- Every request must verify the active business.
- Data must be filtered by business.
- Files must be separated by business.
- AI queries must be restricted to the active business.
- Reports must only use the active business's data.
- Admin access must be logged.
- Subscription access must be controlled by business plan.

Never rely only on frontend filtering for security.

9. SUBSCRIPTION PLANS

Create the following initial plans.

Starter

Target: solo operators

Features:

- One user
- Leads
- Customers
- Quotations
- Invoices
- Jobs
- Calendar
- Basic reports
- Limited AI usage

Growth

Target: small teams

Features:

- Up to five users
- All Starter features
- Employee assignments
- Customer portal
- Automations
- More AI usage
- Email templates
- Advanced reports

Pro

Target: established businesses

Features:

- Up to twenty users
- All Growth features
- Custom roles
- Advanced automations
- API access later
- More storage
- More AI usage
- Priority support

Enterprise

Features:

- Custom users
- Multiple locations
- Custom branding
- Custom integrations
- Dedicated onboarding
- Advanced permissions
- Custom reporting

The first build can use simulated subscription access until payment billing is integrated.

10. SETTINGS

Business Settings

- Business profile
- Logo
- Currency
- Time zone
- Tax
- Document prefixes
- Quotation terms
- Invoice terms
- Default deposit
- Late fees
- Payment instructions
- Email signature
- Notification preferences
- Job statuses
- Lead sources
- Product units
- Customer types

User Settings

- Profile
- Password
- Language
- Notifications
- Appearance
- Active business

AI Settings

- Business description
 - Services offered
 - Service areas
 - Business hours
 - Frequently asked questions
 - Pricing information
 - Deposit policy
 - Warranty policy
 - Tone of voice
 - AI permissions
 - AI usage limits
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11. USER INTERFACE REQUIREMENTS

Design style:

- Professional
- Modern
- Clean
- Easy to understand
- Suitable for nontechnical business owners
- Responsive
- Mobile-first
- Fast
- Accessible

Use:

- Clear navigation
- Large action buttons
- Consistent form layouts
- Status badges
- Confirmation dialogs
- Toast notifications
- Empty states
- Loading states
- Error states
- Search and filtering
- Pagination
- Reusable components
- Consistent spacing and typography

Important user-interface rules:

- Do not overcrowd screens.
- Place the most important actions at the top.
- Use plain language.
- Show totals clearly.
- Make quotation and invoice forms easy to use.
- Make mobile job updates simple for field employees.
- Always show confirmation before destructive actions.
- Autosave long forms where appropriate.

12. TECHNICAL ARCHITECTURE

Build the application using a scalable architecture.

Suggested structure:

Frontend

- Modern React-based web framework
- TypeScript
- Responsive component system
- Server-side rendering where appropriate
- Form validation
- Query caching
- Secure authentication handling

Backend

- TypeScript-based server framework
- REST API or typed RPC architecture
- Modular service structure
- Role-based authorization
- Background job processing
- Validation
- Error logging
- Rate limiting

Database

- PostgreSQL
- Relational schema
- Migrations
- Indexes
- Soft deletion for important business records
- Audit logging

File Storage

- Secure cloud object storage
- Signed URLs
- File-size limits
- File-type validation
- Business-level folder separation

AI

- Large-language-model API
- Structured tool calling
- Retrieval from the business's approved data
- Prompt templates

- AI action logs
- Human confirmation for important actions

Email

- Transactional email provider
- Email templates
- Delivery tracking
- Retry handling

Payments

Add payment gateway integration after the manual payment workflow is complete.

Deployment

- Separate development, staging, and production environments
- Environment variables
- Automated deployment
- Database backups
- Error monitoring
- Application logging
- Uptime monitoring

13. SECURITY REQUIREMENTS

Implement:

- Secure authentication
- Password hashing
- Email verification
- Role-based access control
- Tenant isolation
- Request validation
- Rate limiting
- Cross-site request protection
- Secure file uploads
- Audit logs
- Session expiration
- Account lockout protection
- Data encryption in transit
- Secure environment variables
- Database backups
- Logging of sensitive actions

Sensitive actions include:

- Login
- Password change
- Role change
- Employee invitation
- Customer deletion
- Quotation approval
- Invoice cancellation
- Payment refund
- Data export
- Subscription change
- AI-executed action

Use soft deletion for customers, jobs, quotations, invoices, and payments unless permanent deletion is explicitly required.

14. AUDIT LOGGING

Record:

- User
- Business
- Action
- Record type
- Record identifier
- Previous value
- New value
- Date and time
- IP address where appropriate

Examples:

- Customer created
 - Quotation edited
 - Invoice sent
 - Payment recorded
 - Job status changed
 - Employee role changed
 - File deleted
 - AI action approved
 - AI action rejected
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15. NOTIFICATIONS

Notification types:

- In-app
- Email
- Push notifications later
- SMS later
- WhatsApp later

Initial notifications:

- New lead assigned
- Appointment upcoming
- Quotation approved
- Quotation rejected
- Invoice overdue
- Payment received
- Job assigned
- Task overdue
- Customer message received
- Automation failed

Users should be able to mark notifications as read and configure notification preferences.

16. SEARCH

Create global search across:

- Leads
- Customers
- Quotations
- Jobs
- Invoices
- Payments
- Products
- Documents

Search should support:

- Name
- Telephone number
- Email
- Document number
- Job number

- Address
 - Description
-

17. PDF DOCUMENTS

Generate professional PDFs for:

- Quotations
- Invoices
- Receipts
- Job completion reports
- Customer statements

PDFs should include:

- Company logo
- Business information
- Customer information
- Document number
- Date
- Line items
- Totals
- Payment details
- Terms
- Notes
- Signature areas where appropriate

PDF layouts should be printable on standard paper sizes.

18. MVP BUILD ORDER

Build the application in this exact order.

Sprint 1: Foundation

- Project setup
- Database setup
- Authentication
- Business onboarding
- Multi-tenant architecture
- Roles and permissions
- Main application layout
- Settings

Sprint 2: CRM

- Leads
- Customers
- Customer profiles
- Activities
- Notes
- Search
- Filters

Sprint 3: Products and Quotations

- Product catalogue
- Quotation builder
- Quotation calculations
- PDF generation
- Quotation statuses
- Customer approval link

Sprint 4: Jobs and Calendar

- Job creation
- Convert quotation to job
- Employee assignment
- Tasks
- Checklists
- Calendar
- Appointment reminders

Sprint 5: Invoices and Payments

- Invoice builder
- Convert quotation to invoice
- Record payments
- Partial payments
- Receipts
- Outstanding balances
- Overdue status

Sprint 6: Dashboard and Reports

- Dashboard metrics
- Revenue charts
- Outstanding invoices
- Lead reports
- Quotation reports
- Job reports

- Export functionality

Sprint 7: AI Assistant

- AI chat
- Customer summaries
- Quotation generation
- Message drafting
- Business-data questions
- Confirmation system
- AI audit logs

Sprint 8: Customer Portal

- Secure customer access
- Quotations
- Approvals
- Invoices
- Payments history
- Job status
- Documents
- Messages

Sprint 9: Testing and Launch

- Unit tests
- Integration tests
- Permission tests
- Tenant-isolation tests
- Mobile testing
- Security review
- Performance optimization
- Error handling
- Demo data
- Deployment

19. FEATURES NOT TO BUILD IN THE FIRST MVP

Do not build these until the core application is stable:

- Full accounting system
- Payroll
- Advanced inventory
- Supplier purchasing
- Voice receptionist

- Outbound AI calling
- GPS employee tracking
- Native iOS app
- Native Android app
- Marketplace
- Public API
- Advanced marketing automation
- Social-media scheduler
- Video generation
- Full WhatsApp automation
- Advanced route optimization
- Fleet management
- Multi-location enterprise tools

Prepare the architecture so these can be added later, but do not delay the MVP for them.

20. ACCEPTANCE CRITERIA

The MVP is ready for beta testing when a business owner can:

1. Register and create a business.
 2. Invite an employee.
 3. Add a lead.
 4. Convert a lead into a customer.
 5. Create a product or service.
 6. Create a quotation.
 7. Generate a quotation PDF.
 8. Send or share the quotation.
 9. Allow the customer to approve it online.
 10. Convert the quotation into a job.
 11. Schedule the job.
 12. Assign an employee.
 13. Upload job photos.
 14. Update the job status.
 15. Create an invoice.
 16. Record a deposit or payment.
 17. Generate a receipt.
 18. View revenue and outstanding balances.
 19. Ask the AI assistant questions about business records.
 20. Prevent users from accessing data they do not have permission to view.
 21. Prevent one business from accessing another business's data.
 22. Use the core system on desktop and mobile.
-

21. DEMO BUSINESS

Create a demonstration account for:

Tranquility Windoors and More

Industry:

Aluminum fabrication and installation

Demo products and services:

- Sliding aluminum window
- Louver window
- Aluminum door
- Sliding glass door
- Mesh frame
- Shower enclosure
- Aluminum cupboard
- Window installation
- Door installation
- Site measurement
- Repair service

Demo workflow:

1. Add a customer.
2. Record window measurements.
3. Create a quotation.
4. Require a 70 percent deposit.
5. Approve quotation.
6. Create installation job.
7. Assign installation team.
8. Schedule installation.
9. Upload before-and-after photos.
10. Record final payment.
11. Generate receipt.
12. Mark job completed.

22. AI AGENT DEVELOPMENT INSTRUCTIONS

Follow these rules while building:

1. Build one module at a time.

2. Do not generate the whole application in one uncontrolled step.
 3. Create a task list before coding.
 4. Complete and test each task before moving to the next.
 5. Use reusable components.
 6. Use clear folder structures.
 7. Use strict typing.
 8. Add database migrations.
 9. Add validation to every form and API endpoint.
 10. Add role and tenant checks to every protected action.
 11. Write unit and integration tests.
 12. Do not place secret keys directly in code.
 13. Create an environment-variable example file.
 14. Create demo seed data.
 15. Create setup instructions.
 16. Create deployment instructions.
 17. Create API documentation.
 18. Create a change log.
 19. Log errors clearly.
 20. Never silently ignore failed operations.
 21. Use confirmation dialogs for destructive actions.
 22. Never allow the AI assistant to perform important actions without confirmation.
 23. Keep the application usable throughout development.
 24. Do not begin later-phase features before MVP acceptance criteria are met.
 25. At the end of each sprint, provide:
 26. Completed features
 27. Files created
 28. Database changes
 29. Tests added
 30. Known issues
 31. Next tasks
-

23. FIRST INSTRUCTION TO THE AI CODING AGENT

Begin by completing only the following:

1. Review this entire specification.
2. Create the proposed application architecture.
3. Create the folder structure.
4. Create the database entity relationship plan.
5. Create the MVP development task list.
6. Create the sprint plan.
7. Identify technical risks.

8. Identify security risks.
9. List all required environment variables.
10. Present the plan for approval before writing the full application.

After the architecture is prepared, begin Sprint 1 only.

Do not attempt to build all modules simultaneously.

24. FINAL PRODUCT GOAL

The completed application should allow a service-business owner to run the daily operation of the company from one place.

The business owner should be able to open the application each morning and immediately see:

- Today's jobs
- New leads
- Upcoming appointments
- Quotations awaiting approval
- Overdue invoices
- Payments received
- Employee assignments
- Important tasks
- AI-generated recommendations

The product should feel like an AI business manager, CRM, quotation system, job-management platform, invoicing system, and customer portal combined into one simple application.